# Antitrust Policy and Economic Growth

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ABSTRACT

The potential of being acquired is a significant incentive for start-up firms to perform research and development (R&D); however, mergers and acquisitions (M&A) are also the primary channel by which firms gain market power. This leads to a challenging problem for antitrust policymakers: reduce the misallocation caused by M&A while minimizing the impact on R&D incentives. This paper studies the dynamic antitrust policy problem by developing a novel structural model incorporating heterogeneous firms which can grow either through internal R&D or purchasing competitor firms. Aggregating the individual firms' R&D efforts, the model reveals a dual-edged sword effect of M&A. The absence of M&A reduces the incentives to perform R&D, causing a decrease in the growth rate. However, the lack of antitrust policy leads to highly concentrated markets, decreasing growth. Given this motivation, the paper then examines the optimal antitrust policy problem. Restricting attention to the current method of antitrust enforcement — the premerger notification program — the model shows that significant gains can be made by incorporating dynamic factors into the standard, static antitrust policy tests. However, the model also indicates that a tax on excess profits can be more efficient than the current policy of premerger blocking.

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# 1 Introduction

In 2004, a company named MapQuest dominated the world of online mapping and navigation. At the same time, a small tech company called Where 2 Technologies had also developed a competitor product called Expedition — one that would prove vastly superior to the existing products. Where 2 secured an initial funding deal from a venture capital firm, but any hope of a deal was lost as Yahoo! entered the market the next day. The consensus from venture firms was that it would require too much investment and risk to enable Where 2 to compete with an internet giant like Yahoo. However, one of those VC funds connected Where 2 with another client interested in bringing Where 2's product to the market and out-compete Yahoo. That client was Google and Expedition was re-branded as Google Maps. It is fair to say that, without being acquired, the return on investment for Where 2 Technologies' product would have been scarce.

This story is not just an anecdote of one tech startup. Mergers and Acquisitions (M&A) have become the primary channel for startup companies to receive a return on their investments. Figure 1 displays methods of VC firm exits, the point where venture capital firms exit their investments and receive returns on their investment. It shows that from 2010 to 2019, on average, 78% of VC firms exited via acquisitions compared to 8%, which were exits through public offerings. This trend is a reversal of the VC exit outcomes of the 1990s. From the acquirers' perspective, M&A is a significant channel of investment. Data from SDC platinum, the most comprehensive dataset on U.S. M&A transactions, shows that M&A transaction values, on average, exceed $300 billion a year. However, M&A is a channel rarely considered in the endogenous growth literature. By excluding M&A activity, the literature is missing a significant incentive for investment and a central mechanism for firm growth.

In addition to being a source of significant investment incentives, M&A is also one of the primary methods for firms to consolidate markets and gain monopoly power. Monopolies and antitrust issues have always been a traditional concern of the government. However, recent developments in market consolidation and the rise of large tech companies have brought increased scrutiny. In 2020, the House Judiciary Committee's antitrust panel issued a congressional report following investigation into the big four tech companies: Amazon, Apple, Google and Facebook. In particular, the report raised concerns that the tech companies were no longer confined to their original scope and instead had become multi-headed corporations with arms in many different markets. Another key concern was killer acquisitions, as documented by Cunningham, Ederer and Ma (2021), where a company attempts to maintain their current level of market power by buying a competitor within the same market with the sole purpose of retiring the competing product and shutting down its further development.

From the perspective of an antitrust regulator, fighting markup-induced misallocation while minimizing the distortion on R&D incentives poses a difficult policy problem. As Crandall and Winston (2003) points out, "the substantial and growing challenges of formulating and implementing effective antitrust policies in a new economy characterized by dynamic competition, rapid technological change and important intellectual

property." Given the existing literature, these challenges are difficult to address because the study of antitrust dynamic competition policy is incredibly limited. This sentiment is echoed in the remarks of J. Thomas Rosch, a former commissioner of the Federal Trade Commission:

> If you were to get together a group of antitrust and patent experts, everyone would likely agree broadly that antitrust and intellectual property are complementary in that both areas of law seek to protect and encourage innovation. When it comes to antitrust law, however, promoting innovation is good in theory, but hard in practice.

Thus, the goal of this paper is to help develop a useful tool for the consideration of different dynamic antitrust concerns and policies.

To accomplish that goal, this paper develops a structural model of R&D and M&A to accomplish this task. Firms use R&D investments to both improve the products currently in their portfolio as well as create new products in order to enter into new markets. The M&A block allows the firms to buy the product lines of other firms. This captures M&A's incentive for R&D and the ability for M&A to increase a firm's market power.

After estimating the model on U.S. economic data, the model demonstrates the M&A is a dual edged sword when it comes to economic growth. If there was no M&A, the reduced entry incentives cause a decrease in the economic growth rate. On the other hand, if there was unrestricted M&A, there would also be a decrease in the economic growth rate. This is caused by increased market concentration reducing innovation incentives of incumbant firms.

This paper then examines the impact of antitrust policy on the economy. First, it considers an alternative economy with no antitrust policy. While the growth rate increases in this alternative, the resulting increase in misallocation from market power outweighs the benefit from the increased growth. Next, this paper shows that increased enforcement by the antitrust authority can be welfare improving. However, it also shows that a tax on excise profits can be more effective than the current premeger policy.

The paper is organized as follows. Section 2 lays out the model and Section 3 details how the model is estimated. Section 4 examines how M&A affects growth at both the firm and aggregate levels. Taking the

## 1.1 Literature Review

Despite the significant concern surrounding the dynamic effects of antitrust policy, the literature is incredibly sparse and rarely considers the growth angle. The closest prior research to this paper in goal and approach is Cavenaile, Celik and Tian (2021b). This paper used the Cavenaile et al. (2021b) model as a base and then made several key advancements to increase its applicability. In the Cavenaile et al. (2021b) paper, M&A opportunities arrive exogenously at a common rate for all pairs of firms in a market. The pair of potential merger partners then negotiate bilaterally. Their decision to have a rate

be exogenous results in limited ability of firms to respond to changes in antitrust policy. One advancement of the current paper is to replace the exogenous opportunities with endogenous search. The impact of an exogenous versus endogenous rate is most apparent when considering an antitrust policy shutdown, i.e., all M&A transactions occur freely. In this situation, Cavenaile et al. (2021b)'s model demonstrates that there is only a -0.88% change in the growth rate and a minor impact on market concentration. When running the identical experiment, this paper's model results stand in stark contrast with a 0.5pp decrease in growth and a 51% increases in the number of monopolies, which more realistically reflects both historical changes.

Another significant improvement that this paper makes is to estimate the true U.S. antitrust policy function. Their paper makes the assumption that all transactions above the screening threshold[1] are obstructed with equal probability — an assumption that does not hold true in the empirical data. The consequence of their assumption is to underestimate antitrust enforcement for transactions that result in a high degree of market consolidation since transactions with extremely high HHIs and significant impact on market concentration would have the same probability of being blocked as a transaction that is just over the screening threshold and minimally impacts market concentration. This choice of model assumption would both bias their estimation (the parameter values are chosen to match the distribution of markups of the United States while allowing a higher frequency of large increase in markup transactions) as well as their result that shutting down antitrust enforcement has modest welfare consequences. Other studies have considered specific aspects of dynamic antitrust policy. Fons-Rosen, Roldan-Blanco and Schmitz (2021) focuses on the impact of banning startup acquisitions and finds a positive growth effect. Mermelstein, Nocke, Satterthwaite and Whinston (2020) studies the theory side of dynamic optimal antitrust policy.

A closely related strand of the literature focuses on the interaction of markup and innovation: Peters (2020); Cavenaile, Celik and Tian (2021a); Phillips and Zhdanov (2013); Akcigit, Celik and Greenwood (2016); Cavenaile et al. (2021b); and Pearce and Wu (2022). Peters (2020) shows that R&D activity can have both pro- and anti-competitive effects. Improving a firms existing product lines leads to increasing markups and misallocation while creating new products can displace high-markup incumbents. This paper's model has a very similar finding to Peters (2020). Cavenaile et al. (2021a) found that R&D provides firms with an increased ability to charge markups, resulting in in static welfare losses for consumers. However, they found that the dynamic welfare gains were more significant than the static welfare loss. These findings echo the increasing need for a dynamic antitrust policy.

Additionally, there is a growing strand of the literature on the aggregate impacts of mergers and acquisitions without considering antitrust concerns. This include Jovanovic and Rousseau (2002), Rhodes-Kropf and Robinson (2008), David (2020), Gaillard and Kankanamge (2020), and Bhandari, Martellini and McGrattan (2023). These all show how mergers and acquisitions can improve the improve the allocation of capital. How-

[1]For the latest published guidelines, the screening threshold is an HHI of more than 0.15 and a change in HHI of more than 0.01.

ever, by missing the potential for M&A to cause an increase in market power, it is difficult to determine the true welfare effects.

## 1.2 Institutional Details: U.S. Antitrust Policy

In the United States, antitrust authority is the dual responsibility of the Federal Trade Commission and the Department of Justice, Antitrust Division. While these are two distinct agencies that have different roles and responsibilities, this paper does not reach the level of detail where these differences are significant. As such, in this paper, the two agencies will be referred to as the antitrust authority. The statutory objective of the antitrust authority, found in the Clayton act, is to prevent mergers where

> in any line of commerce or in any activity affecting commerce in any section of the country, the effect of such acquisition may be substantially to lessen competition, or to tend to create a monopoly.

The authority's primary method of antitrust enforcement is the Premerger Notification Program where a proposed M&A transaction[2] is reviewed and, potentially, challenged by the government prior to the transaction[3]. This enforcement program is This process has three primary steps:

1. The transacting parties inform the antitrust authorities of the proposed transaction.

2. The antitrust authority reviews the proposal resulting in either a request for additional information, referred to as a second request, or approval for the transaction to be carried out.

3. For transactions which received a second request, the authority reviews the additional information and determines whether or not to challenge the transaction on antitrust grounds.

A challenge can attempt to completely block a transaction but also commonly requires divestiture of a portion of one of the parties' firm. It is also worth noting that the merging parties can attempt to appeal the challenge in court.

A significant challenge when studying antitrust policy is a lack of transparent quantitative data surrounding the authority's decision-making process. The antitrust authority release two quantitative resources. The first resource is the *Hart-Scott-Rodino Annual Report* (HSR report) in which the Federal Trade Commission and the Department of Justice Antitrust Division are required to report information related to that year's actions for the Premerger Notification Program. Quantitatively, it contains:

[2]Only transactions over a threshold prices are required to be reported in the Premerger Notification Program. This threshold changes by year. At the end fiscal year 2019, the threshold value was $90 million.

[3]While the government also conducts post merger antitrust enforcement (for example, breaking up the Bell system) these actions are incredibly rare

- the number of transactions reported to the antitrust regulator
- the number of requests for second reviews
- the number of transactions that the regulator attempted to block.

The HSR reports lack common concentration information, such as the Herfindahl-Hirschman Index, and only qualitatively describes the reasoning of enforcement actions. Figure 2 plots the second request and challenge rate by time from fiscal year[4] 1996 to fiscal year 2019. This graph shows that second requests are relatively rare and, over the plotted period, an average of 2.3% transaction received any form of challenge by the antitrust authority.

While systematic, quantitative data on the transactions that received second reviews is limited to the HSR reports, a FTC report *Horizontal Merger Investigation Data, Fiscal Years 1996-2011* allows a deeper look into the second review process. The first key data point from the report is the reason the antitrust authority initiated the second review – called the theory of potential violation. Of the identifiable reasons[5], 80% of investigations theory of potential violation concerns horizontal mergers. This fraction is potentially a lower bound since other potential theories include "monopsony theory", "potential competition theory", and "joint venture" which could possibly result from horizontal mergers[6]. Given the antitrust authority's focus on horizontal mergers, this paper only includes horizontal mergers.

The other key data point is Table 3.1 of the report (which is reproduced in this paper as Table 1). This provides the number of second reviews, as well as the number of enforcement actions, by the expected post-merger Herfindahl-Hirschman Index (HHI)[7] and post-merger change in HHI. Panel A of Table 1 counts the number of second reviews. This shows the majority of second review occur at higher levels of HHI and change in HHI. Panel B displays the fraction of second reviews resulting in an enforcement action. Generally, the higher the post-merger HHI and the higher the change in HHI, the higher the likelihood of an enforcement action. However, in the data, this likelihood is not monotonically increasing[8].

[4]The data contained within a year's report covers the U.S. Federal Government's fiscal year of Oct. 1 to Sept. 31.

[5]Non-identifiable reasons are "filing withdrawn by parties during the investigation", "closed after a quick look", and "investigations still open".

[6]All the theory of potential violation other than "vertical theory" comprise 92% of identifiable theory of potential violation

[7]The Herfindahl-Hirschman Index is a commonly used measure of concentration within a market. It is defined as the sum of each market participant's squared market share.

[8]For example, the cell for a 5000-6999 post-merger HHI and 1200-2499 change in HHI has a lower, observed, percentage of enforcement actions than the 4000-4999 post-merger HHI and 1200-2499 change in HHI cell and the 5000-6999 post-merger HHI and 800-1199 change in HHI

# 2 Model

Time is continuous. There is a unit measure of product markets indexed by $i \in [0, 1]$ which are aggregated into a single final good. Within each market, there is a finite set of granular firms which compete strategically and a competitive fringe. These firms can improve their product

## 2.1 Household

There is a single infinitely-lived representative household who discounts the future at rate $\rho$. The household consumes a consumption good $C_t$, with a price $P_t$; supplies one unit of labor to the market at a wage rate $w_t$; and invests in asset $A_t$, with a return $r_t$. Assets evolve as

$$\dot{A}_t = r_t A_t + w_t - P_t C_t.$$

The household determines its consumption, labor supply, and savings to maximize its discounted lifetime utility of:

$$U = \int_0^\infty e^{-\rho t} \log C_t \, \mathrm{d}t. \tag{1}$$

## 2.2 Final Goods

All product markets' goods $y_{i,t}$ are aggregated into a final good $Y_t$ by a competitive final goods firm. This firm employees a Cobb-Douglas aggregator[9]:

$$\log Y_t = \int_0^1 \log y_{i,t} \, \mathrm{d}i. \tag{2}$$

This implies that spending is equalized across all markets:

$$p_{i,t} y_{i,t} = P_t Y_t. \tag{3}$$

## 2.3 Product Markets

Within product market $i$, there is a finite set of strategic firms $F_{i,t}$ and a competitive fringe each producing differentiated varieties of products. For each firm $f \in F_{i,t}$, their output is denoted $y_{f,i,t}$. The fringe's output is $y_{c,i,t}$. Within a product market $i$, the firms' goods are aggregated together into a product market output $y_{i,t}$ by a competitive firm employing C.E.S. technology with elasticity of substitution $\theta > 1$:

$$y_{i,t}^{\frac{\theta-1}{\theta}} = \left( \frac{1}{|F_{i,t}| + 1} \right)^{\frac{1}{\theta}} \left( \sum_{f \in F_{i,t}} y_{f,i,t}^{\frac{\theta-1}{\theta}} + y_{c,i,t}^{\frac{\theta-1}{\theta}} \right). \tag{4}$$

[9] In general, this technology could easily be replaced by a general CES aggregator; however, the unitary elasticity aggregator adds significant clarity during policy analysis. The maximum markup in a market is controlled by the competitive fringe.

The addition of the $\left(\frac{1}{|F_{i,t}|+1}\right)^{\frac{1}{\theta}}$ normalization term remove love for variety from the aggregator. Specifically, the market level output is the same with $n$ firms producing $q$ units of goods or one firm producing $nq$ units of goods. If love for variety was included, the negative consequences of market concentration would be magnified. Demand for each firm's good is given by

$$y_{f,i,t} = \frac{1}{|F_{i,t}|+1} p_{f,i,t}^{-\theta} p_{i,t}^{\theta-1} P_t Y_t \tag{5}$$

where the market price index $p_{i,t}$ is defined by

$$p_{i,t}^{1-\theta} = \frac{1}{|F_{i,t}|+1} \left( \sum_{f \in F_{i,t}} p_{f,i,t}^{1-\theta} + p_{c,i,t}^{1-\theta} \right). \tag{6}$$

## 2.4 Firms

Within the product market, there are two types of firms: a finite set of large strategic firms and a competitive fringe. Each firm is characterized by their productivity $z_{f,t}$. For strategic firms, their productivity is composed of a time-invariant exogenous component $\zeta_f$ and an endogenous component $q_{f,t}$ which evolves over time. For the competitive fringe, their productivity $z_{c,t}$ is assumed to be a fraction $\xi$ of the maximum productivity at time $t$. The state of a market is the vector $\boldsymbol{\zeta}$ and $\boldsymbol{z}_t$. Let $V_{f,t}(\boldsymbol{\zeta}, \boldsymbol{z}_t)$ denote the value of firm $f$ in the market at time $t$. Since the state can be complicated, the full state of the firm will be frequently omitted with a "$\rightarrow$" symbol denoting changes to a firm's state. For example $V_{f,t}(q_{f',t} \rightarrow \lambda q_{f',t})$ denotes that firm $f'$ endogenous productivity changes from $q_{f',t}$ to $\lambda q_{f',t}$.

**Production and Compeition:** To produce their goods, firms combine their productivity linearly with a labor input $h_{f,i,t}$ at a wage rate $w_t$:

$$y_{f,i,t} = z_{f,i,t} h_{f,i,t}. \tag{7}$$

Within each market, the firms in $F_{i,t}$ compete in Cournot competition. For firm $f$ in market $i \in I_{f,t}$, the firm, taking their competitors' prices as given, pick their price, output, and labor input to solve

$$\pi_{f,i,t} = \max_{p_{i,t}, p_{f,i,t}, y_{f,i,t}, h_{f,i,t}} p_{f,i,t} y_{f,i,t} - w_t h_{f,i,t} \tag{8}$$

$$s.t.\ y_{f,i,t} = z_{f,i,t} h_{f,i,t} \tag{9}$$

$$y_{f,i,t} = \frac{1}{|F_{i,t}|+1} p_{f,i,t}^{-\theta} p_{i,t}^{\theta-1} P_t Y_t \tag{10}$$

$$p_{i,t}^{1-\theta} = \frac{1}{|F_{i,t}|+1} \left( \sum_{f \in F_{i,t}} p_{f,i,t}^{1-\theta} + p_{c,i,t}^{1-\theta} \right) \tag{11}$$

$$y_{f,i,t}, h_{f,i,t} \geq 0. \tag{12}$$

The competitive fringe is assumed to set its price to its marginal cost. The resulting prices are pinned down by the nonlinear system

$$s_{f,i,t} = \frac{p_{f,i,t}^{1-\theta}}{\sum_{f' \in F_{i,t}} p_{f',i,t}^{1-\theta} + p_{c,i,t}^{1-\theta}} \tag{13}$$

$$\varepsilon_{f,i,t} = \theta + (1-\theta) s_{f,i,t} \tag{14}$$

$$m_{f,i,t} = \frac{\varepsilon_{f,i,t}}{\varepsilon_{f,i,t} - 1} \tag{15}$$

$$p_{f,i,t} = m_{f,i,t} \frac{w_t}{z_{f,t}} \tag{16}$$

where $s_{f,i,t}$ is firm $f$'s market share in market $i$, $\varepsilon_{f,i,t}$ is firm $f$'s price elasticity, and $m_{f,i,t}$ is firm $f$'s markup over marginal cost. After computing prices, output is determined by (10) and the labor input is determined by (7). Since $\theta > 1$, market share is decreasing in price. A firm that has higher productivity is able to charge a lower price capturing a larger share of the market. As the firm captures more of the market share, its product becomes more inelastic allowing it to charge a higher markup and earn higher monopoly profits.

**Research and Development:** To increase their profitability, firms can perform R&D to improve the productivity of their current products. Each good's productivity sits on a quality ladder with step size $\lambda > 1$. Each time a firm improves its product, the good's productivity takes a single step up the ladder — its productivity goes from $q_{f,i,t}$ to $\lambda q_{f,i,t}$. The resulting change in firm $f$'s value is

$$V_{f,t}\left(q_{f,i,t} \to \lambda q_{f,i,t}\right) - V_{f,t}. \tag{17}$$

Improvements arrive to the firm exogenously according to a Poisson arrival process at an endogenous innovation rate $j_{f,i,t}$. The cost of achieving that rate is $c_j\left(j_{f,i,t}\right) Y_t$ units of goods where $c_j$ is assumed to be strictly convex. The firm chooses its innovation rate to solve

$$\max_{j_{f,i,t} \geq 0} \left\{ j_{f,i,t} \left( V_{f,t}\left(q_{f,i,t} \to \lambda q_{f,i,t}\right) - V_{f,t} \right) - c_j\left(j_{f,i,t}\right) Y_t \right\}. \tag{18}$$

A product's productivity never decreases.

**Mergers and Acquisitions:** The second source of firm growth is through mergers and acquisitions. M&A transactions are between current market incumbents and for single market's product. While M&A transactions are traditionally thought of as one company buying the entirety of another, it is also frequently observed in the data that one company buys only a portion of another[10]. Within market $i$, firm $f \in F_{i,t}$ receives an opportunity to attempt to purchase firm $f' \in F_{i,t}$ market $i$ product at rate $a_{f',f,i,t}$. The potential buyer endogenously sets this rate $a_{f',f,i,t}$ at a cost $c_a\left(a_{f',f,i,t}\right) Y_t$. When

[10]For example, Google purchased only Motorola's cell phone division.

the parties agree to undertake a transaction, the prices is set by Nash bargaining where the buyers bargaining power is $\beta$. After the transaction is complete, the selling party exits the market. Crucially, from a competition perspective, that means there is one less competitor in the market.

After the buyer acquires the seller's product, it has two options. The first option is the buyer can either combine its product with the seller's product to product a single, potentially, improved product. The second option is that the buyer can shelve the seller's product effectively killing it. This second option results in no change to the buyer's productivity within the market. Combining the two products is not free. After the seller's product is transferred to the buyer, the buyer observes the combination cost $\varepsilon_{f',f,i,t}$ which is drawn from an exponential distribution with shape parameter $\omega$. If the buyer chooses to combine the products, the resulting productivity $\tilde{q}_{f,i,t}$ is

$$\tilde{q}_{f,i,t} = \zeta q_{f,i,t}^{\alpha} q_{f',i,t}^{1-\alpha}. \tag{19}$$

If the buyer chooses to shelve the seller's product, then the buyer's product's productivity remains unchanged.

Prior to every M&A transaction, the antitrust authority reviews the transaction and can choose to block the transaction. In the baseline model, the antitrust authority only considers the post-merger HHI and the change in HHI[11]. Let $\text{hhi}_{f',f,i,t}$ denote the post-merger HHI of firm $f$ buying the market $i$ good of firm $f'$ at time $t$ and $\Delta\text{hhi}_{f',f,i,t}$ denote the resulting change in HHI. The antitrust authority blocks a transaction with probability $B_{f',f,i,t} = B\left(\text{hhi}_{f',f,i,t}, \Delta\text{hhi}_{f',f,i,t}\right)$. This function is an input to the model and is estimated in Section 3.1. If the regulator blocks a transaction, the parties do not have a chance to restructure the transaction and the deal is dead.

The total surplus for firm $f$ buying the good of firm $f'$ in market $i$ at time $t$ and combining the goods, denoted $TS^{C}_{f',f,i,t}$, is

$$\begin{aligned} TS^{C}_{f',f,i,t} = {} & V_{f,t}\left((q_{f,i,t}, q_{f',i,t}) \to \left(\zeta q_{f,i,t}^{\alpha} q_{f',i,t}^{1-\alpha}, 0\right)\right) \\ & + V_{f',t}\left((q_{f,i,t}, q_{f',i,t}) \to \left(\zeta q_{f,i,t}^{\alpha} q_{f',i,t}^{1-\alpha}, 0\right)\right) \\ & - V_{f,t} - V_{f',t} - \varepsilon_{f',f,i,t}. \end{aligned} \tag{20}$$

The total surplus for the same transaction where the buyer kills the product, denoted $TS^{K}_{f',f,i,t}$, is

$$TS^{K}_{f',f,i,t} = V_{f,t}\left(q_{f',i,t} \to 0\right) + V_{f',t}\left(q_{f',i,t} \to 0\right) - V_{f,t} - V_{f',t}. \tag{21}$$

Given the Nash bargaining structure of M&A transactions, the buyer's surpluses are defined as

$$BS^{C}_{f',f,i,t} = \beta TS^{C}_{f',f,i,t} \tag{22}$$

$$BS^{K}_{f',f,i,t} = \beta TS^{K}_{f',f,i,t}. \tag{23}$$

[11] This assumption allows the model to take full advantage of the limited, publicly-available data described in Section 1.2.

Define firm $f$'s gain from combining $\Delta BS^C_{f',f,i,t}$ as

$$\Delta BS^C_{f',f,i,t} = BS^C_{f',f,i,t} - BS^K_{f',f,i,t}. \tag{24}$$

The probability of the firm combining $P^C_{f',f,i,t}$ is

$$P^C_{f',f,i,t} = 1 - \exp\left(-\omega \max\left\{0, \Delta BS^C_{f',f,i,t}\right\}\right). \tag{25}$$

Thus the expected buyer's surplus $\mathbb{E}\left[BS_{f',f,i,t}\right]$ is

$$\mathbb{E}\left[BS_{f',f,i,t}\right] = \mathbb{E}\left[\left(1 - B_{f',f,i,t}\right)\left(P^C_{f',f,i,t} BS^C_{f',f,i,t} + (1 - P^C_{f',f,i,t}) BS^K_{f',f,i,t}\right)\right]. \tag{26}$$

When the buyer decides its $a_{f',f,i,t}$, it solves

$$\max_{a_{f',f,i,t}} \left\{a_{f',f,i,t} \mathbb{E}\left[BS_{f',f,i,t}\right] - c_A\left(a_{f',f,i,t}\right) Y_t\right\}. \tag{27}$$

To simplify notation, let $MA_{f',f,i,t}\left(\varepsilon_{f',f,i,t}\right)$ denote the change of state when firm $f$ buys the good of firm $f'$ in market $i$ where the combination cost is $\varepsilon_{f',f,i,t}$.

**Entry:** Creating a new product through expansion results in the firm expanding into a new market. This research effort is undirected meaning the firm cannot target which market to enter. When they develop a new product, the firm receives a uniform random draw of which market $i \in [0,1]$ it enters. Upon entering the new market, the productivity of the new good is a step above the maximum good-specific productivity currently existing in that market. Let $q_{\ell,i,t} = \max\left\{q_{f,i,t} : f \in F_{i,t}\right\}$. The new product's productivity is $\lambda q_{\ell,i,t}$. This is intended to capture new products being superior versions of existing products. The expected value to the firm of a new product is

$$\mathbb{E}_{i'}\left[V\left(\left\{q_{f,i,t}\right\}_{i \in I_{f,t}} \to \left\{q_{f,i,t}\right\}_{i \in I_{f,t}} \cup \left\{q_{f,i',t}\right\}\right)\right] - V.$$

New products arrive to the firm according to a Poisson arrival process at an endogenous expansion rate $x_{f,t}$. The cost of achieving that rate is $c_x\left(x_{f,t}\right) Q_t$ units of goods where $c_j$ is assumed to be strictly convex and differentiable. The firm chooses its expansion rate to satisfy

$$\max_{x_{f,t} \geq 0} \left\{x_{f,t}\left(\mathbb{E}_{i'}\left[V\left(\left\{q_{f,i,t}\right\}_{i \in I_{f,t}} \to \left\{q_{f,i,t}\right\}_{i \in I_{f,t}} \cup \left\{q_{f,i',t}\right\}\right)\right] - V\right) - c_x\left(x_{f,t}\right) Q_t\right\}. \tag{28}$$

Within each product market, there is a maximum size cap of $\bar{F}$ firms. If the market is drawn, and it already has $\bar{F}$ firms, then the firm with the least profitable product[12], in terms of sales minus wages paid to production workers.

[12] If multiple firms are tied for least profitable, one is picked with equal probability to exit the market.

**Firm's Value Function:** Combing the previous description results in the definition of the firm's value function of

$$
\begin{aligned}
r_t V_{f,t} - \dot{V}_{f,t} = & \underbrace{\pi_{f,t}}_{\text{Current Gross Profits}} \\
& + \underbrace{\max_{j_{f,i,t}} \left\{ j_{f,i,t} \left( V_{f,t} \left( q_{f,i,t} \to \lambda q_{f,i,t} \right) - V_{f,t} \right) - c_J \left( j_{f,i,t} \right) Y_t \right\}}_{\text{Firm's R\&D Improvement}} \\
& + \underbrace{\sum_{\substack{f' \in F_{i,t} \\ f' \neq f}} j_{f',i,t} \left( V_{f,t} \left( q_{f',i,t} \to \lambda q_{f',i,t} \right) - V_{f,t} \right)}_{\text{Competitors' R\&D Improvement}} \\
& + \underbrace{\sum_{z} x_{z,t} \left( V_{f,t} \left( \{ q_{f',i,t} \} \to \{ q_{f',i,t} \} \cup q_{f'',i,t} \right) - V_{f,t} \right)}_{\text{New Compeitor's Entry}} \\
& + \underbrace{\sum_{\substack{f' \in F_{i,t} \\ f' \neq f}} \max_{a_{f',f,i,t}} \left\{ a_{f',f,i,t} \mathbb{E} \left[ BS_{f',f,i,t} \right] - c_A \left( a_{f',f,i,t} \right) Y_t \right\}}_{\text{Own Acquisitions}} \\
& + \underbrace{\sum_{\substack{f' \in F_{i,t} \\ f' \neq f}} a_{f,f',i,t} \mathbb{E} \left[ SS_{f',f,i,t} \right]}_{\text{Being Acquired}} \\
& + \underbrace{\sum_{\substack{f', f'' \in F_{i,t} \\ f', f'' \neq f \\ f' \neq f''}} a_{f'',f',i,t} \left( \left( 1 - B_{f'',f',i,t} \right) \mathbb{E} \left[ V_{f,t} \left( MA_{f'',f',i,t} \left( \varepsilon_{f'',f',i,t} \right) \right) \right] - V_{f,t} \right)}_{\text{Competitors' M\&A Actions}}
\end{aligned}
\tag{29}
$$

**Market Clearing:** Final goods are split among consumption and investment by

$$
Y_t = C_t + \int_0^1 \sum_{f \in F_{i,t}} c_J(j_{f,i,t}) Y_t + c_a(a_{f,i,t}) Y_t \, \mathrm{d}i + \int_0^m c_X(x_{f,t}) Y_t \, \mathrm{d}f. \tag{30}
$$

The aggregate entry rate $X_{z,t}$ must match the aggregated individual expansion rates

$$
X_{z,t} = \int_{\{f : z_f = z\}} x_{f,t} \, \mathrm{d}f. \tag{31}
$$

Within the labor market, markets clear when aggregate labor demand equals one. That is

$$
1 = \int_0^1 \sum_{f \in F_{i,t}} h_{f,i,t} + h_{c,i,t} \, \mathrm{d}i. \tag{32}
$$

Finally, household assets must equal the value of the economy's firms:

$$A_t = \int_0^m V_{f,t} \, \mathrm{d}f \tag{33}$$

## 2.5 Equilibrium

This paper focuses on the impact of policy on the balanced growth path.

**Definition 1.** *A **balanced growth path** is a set of firm level allocations*

$$\left\{ x_{f,t}, \left\{ y_{f,i,t}, h_{f,i,t}, j_{f,i,t}, \{a_{f',f,i,t}\}_{f' \in F_{i,t}, f' \neq f} \right\}, y_{c,i,t}, h_{c,i,t} \right\}_{i,t},$$

*market distribution $\mu_t$, aggregate allocations $\{C_t, A_t, Y_t\}_t$, entry rate $X_{z,t}$, and prices $\left\{ P_t, r_t, w_t, \{p_{f,i,t}\}_{f,i,t} \right\}$ such that*

1. *Consumption $C_t$ and assets $A_t$ maximize household welfare (**??**).*
2. *The firm allocation solve the optimization problems in (**??**) and (**??**).*
3. *The distribution of markets $\mu_t$ satisfies (**??**) and $\dot{\mu}_t = 0$.*
4. *The market clearing conditions (30)-(33) all hold.*
5. *Aggregate output, consumption, wages, investment, and productivity all grow at the same rate $g$.*

Part 5 of the definition implies

$$Y_t = Y e^{gt} \tag{34}$$
$$C_t = C e^{gt} \tag{35}$$
$$w_t = w e^{gt} \tag{36}$$
$$Q_t = Q e^{gt}. \tag{37}$$

## 2.6 Equilibrium Results

On the balanced growth path, there are several analytic results that can be derived.

**Interest Rates:** The interest rate is given by

$$r_t = \rho + g. \tag{38}$$

**Welfare:** On the balanced growth path, household welfare is

$$U = \frac{1}{\rho} \log C + \frac{1}{\rho^2} g.$$

**Prices and Production:** Given that within-market competition is assumed to follow Cournot game, prices are pinned down via a nonlinear system

$$s_{f,i,t} = \frac{p_{f,i,t}^{1-\theta}}{\sum_{f' \in F_{i,t}} p_{f',i,t}^{1-\theta} + p_{c,i,t}^{1-\theta}} \tag{39}$$

$$\varepsilon_{f,i,t} = \theta + (1-\theta) s_{f,i,t} \tag{40}$$

$$p_{f,i,t} = \frac{\varepsilon_{f,i,t}}{\varepsilon_{f,i,t} - 1} \frac{w_t}{z_{f,t} q_{f,i,t}} \tag{41}$$

where $s_{f,i,t}$ is firm $f$'s market share in market $i$ and $\varepsilon_{f,i,t}$ is firm $f$'s market $i$ elasticity. After computing prices, output is determined by (10) and the labor input is determined by (7). To simplify things going forward, define

$$\begin{aligned}
q_{i,t} &= \max\{q_{f,i,t} : f \in F_{i,t}\} \\
z_{i,t} &= \max\{z_{f,t} : f \in F_{i,t}\} \\
\tilde{q}_{f,i,t} &= \frac{q_{f,i,t}}{q_{i,t}} \\
\tilde{z}_{f,i,t} &= \frac{z_{f,t}}{z_{i,t}} \\
m_{f,i,t} &= \frac{\varepsilon_{f,i,t}}{1 - \varepsilon_{f,i,t}} \\
m_{i,t} &= \left( \sum_{f \in F_{i,t}} \left( \frac{m_{f,i,t}}{\tilde{z}_{f,i,t} \tilde{q}_{f,i,t}} \right)^{1-\theta} + \left( \frac{\eta}{z_{i,t}} \right)^{1-\theta} \right)^{\frac{1}{1-\theta}}.
\end{aligned}$$

**Aggregate Output:** Aggregate output can be expressed as

$$Y_t = Q_t Z_t \mathcal{M}_t \tag{42}$$

where the goods' productivity index $Q_t$ is defined by

$$\log Q_t = \int_0^1 \log q_{i,t} \, \mathrm{d}i, \tag{43}$$

the firms' productivity index $Z_t$ is defined by

$$\log Z_t = \int_0^1 \log z_{i,t} \, \mathrm{d}i, \tag{44}$$

and a mark up term $\mathcal{M}_t$ defined by

$$\mathcal{M}_t = \frac{\exp\left( \int_0^1 \log m_{i,t}^{-1} \, \mathrm{d}i \right)}{\int_0^1 \left( \sum_{f \in F_{i,t}} m_{f,i,t}^{-\theta} \tilde{z}_{f,i,t}^{\theta-1} \tilde{q}_{f,i,t}^{\theta-1} + \left( \frac{\eta}{z_{i,t}} \right)^{\theta-1} \right) m_{i,t}^{\theta-1} \, \mathrm{d}i}. \tag{45}$$

**Growth:** On the balanced growth rate, the growth rate $g$ is given by

$$g = \underbrace{\log(1+\lambda)(J+X)}_{\text{Improvments and Entry}} + \underbrace{\int_0^1 \sum_{\substack{f',f\in F_{i,t}\\ f\neq f'}} a_{f',f,i,t}(1-B_{f',f,i,t})\mathbb{E}\left[\log\left(\frac{q^{MA}_{\ell,f',f,i,t}}{q_{\ell,i,t}}\right)\right] \mathrm{d}i}_{\text{M\&A}} . \tag{46}$$

where

$$J = \int_0^1 \sum_{\substack{f\in F_{i,t}\\ q_{f,i,t}=q_{i,t}}} j_{f,i,t}\,\mathrm{d}i \tag{47}$$

$$X = \sum_z X_{z,t} \tag{48}$$

and $q^{MA}_{\ell,f',f,i,t}$ is the market leader's good's productivity after firm $f$ acquires the good of firm $f'$. For the M&A term, it is important to note that $\log\left(\frac{q^{MA}_{\ell,f',f,i,t}}{q_{\ell,i,t}}\right)$ can either be positive or negative. The M&A term also reveals the direct effect of antitrust policy on growth — blocking transactions where $\log\left(\frac{q^{MA}_{\ell,f',f,i,t}}{q_{i,t}}\right) > 0$ directly reduces the growth rate.

# 3 Estimation

## 3.1 The Antitrust Policy Function

One of the crucial inputs to the model is the probability of antitrust enforcement $\mathbb{P}(B|\text{hhi},\Delta\text{hhi})$. Given the limited publicly available data on antitrust decision-making, estimating this function is challenging. An approach commonly taken by the existing literature[13] is to assume all transactions with a post-merger HHI and change in HHI above the horizontal merger guidelines receive enforcement with a constant probability. However, as documented in Section 1.2, the probability of a transaction receiving an enforcement action increases with the post-merger HHI and change in HHI. This paper's approach is to decompose the conditional probability function into two parts — one which can be directly estimated from the publicly available data and the other which will be estimated as part of the moment matching exercise.

The law of total probabilities allows $\mathbb{P}(B|\text{hhi},\Delta\text{hhi})$ to be written as

$$\mathbb{P}(B|\text{hhi},\Delta\text{hhi}) = \mathbb{P}(B|S,\text{hhi},\Delta\text{hhi})\,\mathbb{P}(S|\text{hhi},\Delta\text{hhi}) + \mathbb{P}\left(B|\overline{S},\text{hhi},\Delta\text{hhi}\right)\mathbb{P}\left(\overline{S}|\text{hhi},\Delta\text{hhi}\right). \tag{49}$$

[13] See Cavenaile et al. (2021b).

Since the first step in the antitrust review process is a second review, $\mathbb{P}\left(B|\overline{S}, \text{hhi}, \Delta\text{hhi}\right) = 0$. Thus, it follows that (49) becomes

$$\mathbb{P}\left(B|\text{hhi}, \Delta\text{hhi}\right) = \mathbb{P}\left(B|S, \text{hhi}, \Delta\text{hhi}\right)\mathbb{P}\left(S|\text{hhi}, \Delta\text{hhi}\right). \tag{50}$$

This transforms the conditional probability of blocking into a combination of the conditional probability of a second review and the conditional probability of blocking given a second review. The remainder of this section will detail how each object is estimated.

Beginning with the conditional blocking probability given a second review $\mathbb{P}\left(B|S, \text{hhi}, \Delta\text{hhi}\right)$, Table 3.1 of *Horizontal Mergers Investigation Data, Fiscal Years 1996-2011* brakes down the fraction of second reviews by U.S. antitrust authorities that received enforcement action by post-merger HHI and change in HHI. Performing logistic regression[14] results in an estimate of $\mathbb{P}\left(B|S, \text{hhi}, \Delta\text{hhi}\right)$ given by

$$\text{logit}\,\hat{\mathbb{P}}\left(B|S, \text{hhi}, \Delta\text{hhi}\right) = -0.21 + 1.86\,\text{hhi} + 0.1862\,\Delta\text{hhi} + 10.34\,\text{hhi} \times \Delta\text{hhi}. \tag{51}$$

To evaluate the accuracy of this classifier, twenty percent of the data was withheld from training. For the test, a transaction is considered blocked if $\hat{\mathbb{P}}\left(B|S, \text{hhi}, \Delta\text{hhi}\right) > \frac{1}{2}$. The resulting precision[15] is 0.80 and recall is 0.93. Thus, the estimate is relatively accurate on the test data.

The second term in (50), the conditional probability of a second review, can not be directly estimated from the data. While *Horizontal Mergers Investigation Data, Fiscal Years 1996-2011* does have the number of second reviews by post-merger HHI and change in HHI, there is no data source that provides the number of transactions that did not receive a second review by post-merger HHI and change in HHI. To alleviate this challenge, this paper adopts an approach that enables indirect estimation of a second review policy function. Consider the antitrust policy maker's decision problem to issue a request for a second review. Second reviews are not free for the antitrust policy maker: they take a substantial amount of staff work to review the information and issue a decision. Thus an optimal antitrust policymaker only issue second-review requests to transactions that they believe they are likely to block after the second review.

The policy maker's second-review problem can be modeled as a discrete choice problem. Out of the set of all M&A transactions, the policy maker needs to pick which set of transactions receives a second review subject to the budget it has allocated for antitrust enforcement. The antitrust authority orders transactions by blocking likelihood and picks transactions until it exhausts its budget. This results in a second review threshold $\bar{s}$ where any transaction with

$$\hat{\mathbb{P}}\left(B|S, \text{hhi}, \Delta\text{hhi}\right) \geq \bar{s} \tag{52}$$

receives a second review and every other transaction passes unopposed. This threshold, $\bar{s}$, becomes a model parameter that will be indirectly estimated.

[14] More details on this regression are provided in Appendix **??**

[15] Precision and recall are common accuracy measures for classification methods. Here, precision is the fraction of transactions predicted to be blocked that were blocked in the test data. Recall is the fraction of transactions blocked in the test data predicted to be blocked.

## 3.2 Functional Forms

The model admits arbitrary functional forms on the firm's cost functions. This paper will use simple, convex-cost functions

$$c_j\left(j_{f,i,t}\right) = \varphi_j j_{f,i,t}^{1+\gamma_j} \tag{53}$$

$$c_x\left(x_{f,t}\right) = \varphi_x x_{f,t}^{1+\gamma_x} \tag{54}$$

$$c_a\left(a_{f',f',i,t}\right) = \varphi_a a_{f',f',i,t}^{1+\gamma_a}. \tag{55}$$

## 3.3 Parameter Estimation

In addition to second review outcome probability $B\left(\text{hhi}, \Delta\text{hhi}|S\right)$, the model has a total of twenty parameters: $\alpha$, $\beta$, $\eta$, $\gamma_a$, $\gamma_j$, $\gamma_x$, $\lambda$, $m$, $\nu$, $\omega$, $\varphi_a$, $\varphi_j$, $\varphi_x$, $\psi$, $\theta$, $\zeta$, $z_1$, $z_3$, and $\bar{s}$. The discount rate $\rho$ and labor disutility parameters $\nu$ and $\psi$ are set to standard values of $\rho = 0.05$, $\nu = 24$, and $\psi = 2$. Given the difficulty in credibly identify R&D curvature parameters without exogenous R&D cost variation, this paper follows Acemoglu, Akcigit, Alp, Bloom and Kerr (2018)'s suggestion of using $\gamma_x = \gamma_j = 1$. For a similar reason, this paper sets $\gamma_a = 1$. The remaining fifteen parameters are estimated using the method of simulated moments. The remaining portion of this subsection will describe the target moments and their data sources. The data underlying each moment is restricted to 1996-2019.

To capture the U.S.'s growth dynamics, the model is estimated to match the average National Income and Product Accounts' GDP growth rate of 2.41% as well as the ratio of R&D investments to GDP of 0.04. Given the paper's interest, it is critical to accurately depict the interaction of competition and research effort. For almost all parameter sets, this model will produce an inverted-U relationship between level of competition and research intensity as documented by Aghion, Bloom, Blundell, Griffith and Howitt (2005). The goal is to match the inverted-U shape in the data. Following Cavenaile et al. (2021a), the model is estimated to match a quadratic regression of relative sales on the log of RD spending[16]. The data underlying this regression is Compustat firm-years with a positive R&D expenditure. Sales share is defined by the shares of sales within its four-digit SIC. The result of this regression is

$$\log \text{RD}_{f,t} = 1.39 \times \text{share}_{f,t} - 1.11 \times \text{share}_{f,t}^2 + \varepsilon_{f,t}. \tag{56}$$

The next set of moments are statistics on observed M&A transactions where the acquirer was a U.S. publicly traded corporation. The data source is SDC platinum which contains records of individual M&A transactions. Barnes, L. Harp and Oler (2014) reports that is generally reliable and complete source of the target transactions. The first moment is the aggregate M&A transaction value to GDP of approximately 5% per year. Next

[16] In Cavenaile et al. (2021a), they used patent counts and improvement arrival rates. Given the multi-product nature of this model, R&D spending adds up to a firm-level variable more easily than the arrival rates.

is the mean and standard deviation of merger premium. In order to discipline the killer acquisition rate, the model is targeted to matchCunningham et al. (2021)'s lower bound on the estimated killer acquisition rate of 5.3%. The final M&A moment is the antitrust authority's second request rate of 3%. The final set of moments are related to the firm's productivity distribution.

These data moments are summarized in Panel B of Table 2. The parameters are chosen to match these target moments. The resulting parameters are shown in Panel A of Table 2 and the resulting model moments are shown in Panel B. While the model is not exactly identified, the simulated moments are close enough that the results of this parameter set should be reflective of the exactly identified model.

# 4 The Effects of M&A on Growth

The goal of this section is to understand the impact of M&A on R&D effort. The clearest way to examine the impact is to compare the model with an alternative where the antitrust regulator blocks all M&A. The next section examines the impact at the level of the individual firms. The following section examines the aggregate impact and the associated impact on growth.

## 4.1 Impact of M&A Across the Firm Distribution

To begin, consider the entry effort margin. Table 3 shows the firm-level entry effort in both the baseline model and the no M&A case by productivity level. Without M&A, every firm decreases their entry effort showing that M&A and entry are compliments. The the 53% decrease for the low productivity firms is the largest response. The reason is simple: removing M&A is removing a channel for firms to receive a return on their entry investment. While an individual, low productivity firm's entry effort is relatively small, they make up 50% of all firms. Thus, they comprise 36% of the aggregate entry rate[17]. Even the high productivity firm experiences a negative decline in their entry rate when there is no M&A. For the high productivity firms, the removal of the investment return channel of being bought out is part of the story. However, their profits are also effected by the reduced ability to consolidate markets[18].

After seeing how the entry margin of R&D decreased without the ability to perform M&A, a natural question is if the improvement margin is similarly impacted? Figure **??** displays the inverse cumulative distribution function of the change in improvement rate when M&A is removed[19]. It reveals that approximately 75% of divisions would perform less improvement R&D in a world without M&A. This results in around a 19% decrease in

[17]The middle productivity firm comprise 46% of the aggregate entry rate and the high productivity firm comprise the remaining 18%.

[18]A simple measure of this is the number of monopoly and duopoly markets. In the baseline model, 24% of markets are monopolies or duopolies. Without M&A, it is nearly zero markets.

[19]The distribution of divisions underlying this figure is the distribution from the baseline model

the average improvement effort[20]. However, in addition to the changes in improvement rates, the distribution of divisions also changes between the two cases. The lack of M&A results in markets being more competitive. Given the model matches research and development's inverted U-shape relationship with competition[21], this increase in competitiveness results in a significant increase in average improvement effort compared to the baseline.

### 4.2 M&A and Aggregate Growth

Aggregate growth can be decomposed into four components: improvement $g_{\text{Improvement}}$, entry $g_{\text{Entry}}$, M&A combination $g_{\text{Combine}}$, and M&A kill $g_{\text{Kill}}$. Table 4 decomposes the aggregate growth rate into this four components. The largest contributor, adding 1.29pp to the aggregate growth rate, is the entry component. Next is the improvement component adding 0.69pp. M&A resulting in combining products contributes 0.41pp. Some what surprisingly, M&A transactions resulting in the acquirer killing the acquired product only reduces the aggregate growth rate by 0.01pp.

To understand the impact of M&A on these aggregate components, consider the alternative model where all M&A is banned. The aggregate growth rate in this case is 1.9% which is 0.49pp lower than the baseline model. The decomposition into the four components, as well as the difference between the baseline and the no M&A case, is also shown in Table 4. Consistent with the previous section, the improvement component does increase by 0.26pp. However, this does not make up for the -0.35pp change in the entry rate component. This also reveals that the M&A substitutes with aggregate improvement R&D but is complementary with aggregate entry R&D. However, as the next section will show, the presence of M&A does increase aggregate misallocation.

## 5 Competition Policy

### 5.1 The Dual Edge Sword Effect of M&A

The previous section showed that banning all M&A would reduce the aggregate growth rate by nearly a half percentage point. Given that M&A both acts as an incentive for entry as well as a source of growth through the combining products, a reasonable hypothesis is that allowing all M&A transactions may increase growth. As Table 5 reveals, the growth rate is 0.24pp higher than the baseline case. However, it also reveals that the household would prefer not live in either the economy with no M&A or the economy with no enforcement. The difference in welfare from the baseline economy is determined by two things: the change in the economic growth rate and the change in static misallocation. In the case where there is no M&A, the impact of the reduced misallocation does not make up for the reduced growth rate. In the case where there is

[20]Computing the average based of the baseline distribution of divisions.
[21]Documented in Aghion et al. (2005).

no antitrust enforcement, the increased growth rate does not offset the increased static misallocation. The source of this increased misallocation is increased concentration in markets. For example, the fraction of monopoly markets rises from less than one percent in the no M&A case to approximately 51% in the no enforcement case[22]. This increase in concentration reduces the level base level of consumption, $C$, by 24% compared to the baseline model. In both cases, the household's welfare decreases. The no M&A case has a decrease of 18.15% and the no enforcement case has a decrease of 7.86%. This shows that an optimal antitrust authority must consider its impact on growth in addition to its impact on static misallocation.

## 5.2 The Antitrust Enforcement Rate

An important parameter in the model's antitrust rule is the threshold $\bar{s}$. The parameter controlled which transactions in the model received second reviews. This parameter was targeted to match the second request rate of the United State's antitrust authority. This subsection will investigate the impact of different rates of antitrust enforcement by varying $\bar{s}$. When $\bar{s} = 0$, every transaction receives a second review by the antitrust authority. When $\bar{s} = 1$, no transactions receive a second review[23]. Figure **??** plots key economic aggregates for different values of $\bar{s}$ from 0 to 1. Panel A shows the change in household welfare compared to the estimated level of $\bar{s} = 0.82$. The welfare curve is parabolic. The reason for the declining welfare follows from the previous subsection[24]. Panel B shows that the economic growth rate is increasing in $\bar{s}$ meaning, as the enforcement rate decreases, the growth rate increases. Output has the opposite relationship with $\bar{s}$. Panel C shows the the base level of output $Y$ as a function of $\bar{s}$. Panel D reveals the cause of the declining output — the average level of markup increases as $\bar{s}$ increases.

Panel A of Figure **??** also reveals that an increased level of enforcement setting $\bar{s} = \boldsymbol{xxx}$ maximizes household welfare. The results in a $\boldsymbol{xxx}$% increase in household welfare. This increased level of enforcement would result in a $\boldsymbol{xxx}$% increase in the number of antitrust reviews and a $\boldsymbol{xxx}$% increase in the number of enforcement actions. It is difficult to estimate the added cost of an increased enforcement rate; however, it is likely the added cost would not outweigh the **xxx**% increase in household welfare.

## 5.3 Simple Output Rules

While the predictive analysis to identify anticompetitive mergers in the real world is extremely difficult, within the model, it is simple. Given the unitary elasticity final-good's aggregator, market level revenue $p_{i,t}y_{i,t}$ is equal across markets. If a merger results in a lessening of competition, it also results in a decrease in output. So, within

[22]In the baseline model, approximately 4% of markets are monopolies.

[23]This is equivalent to having no antitrust enforcement

[24]The case of $\bar{s} = 1$ is the same as the case where there is no M&A enforcement. The case of $\bar{s} = 0$ still allows for M&A but adds a significant amount of blocking compared to the estimated $\bar{s} = 0.82$.

the model, a simple rule could reject a proposed merger if it results in a decrease in some measure of output.

From a purely static perspective, this rule blocks the merger if it the post-merger level of market output less than the premerger level of market output. Let $\mathbb{E}y^{M}_{f',f,i,t}A$ be the expected level of output after the merger[25]. The pre-merger level of output is $y_{i,t}$. So, the simple static rule is to block a merger if

$$\mathbb{E}y^{MA}_{f',f,i,t} < y_{i,t}. \tag{57}$$

The static nature of this rule means it ignores the possibility that the relationship may reverse at some future date. It is possible that $\mathbb{E}y^{M}_{f',f,i,t+s}A > \mathbb{E}y_{i,t+s}$ for some $s > 0$ and remains that way. A dynamic alternative to the static simple rule to to compare the, discounted, average future output paths. Let

$$E^{MA}_{f',f,i,t} = \mathbb{E}\int_0^\infty e^{-\rho s} y^{MA}_{f',f,i,t+s}\,\mathrm{d}s \tag{58}$$

$$E_{i,t} = \mathbb{E}\int_0^\infty e^{-\rho s} y_{i,t+s}\,\mathrm{d}s. \tag{59}$$

The equivalent dynamic rule to (57) is block a proposed merger if

$$E^{MA}_{f',f,i,t} < E_{i,t}. \tag{60}$$

The main difference between the two rules is that rule (60) accounts for the fact that the post-merger output may grow faster than the pre-merger path. The results of these two proposed policies is shown in Table 5.

**TODO**

## 5.4 Comparing to a Tax on Excess Profits

Several authors, such as Avi-Yonah (2020), François, Oliveira, Planterose and Zucman (2022), and Avi-Yonah (2022), propose utilizing a tax on excess profits to combat different sources of economic rents. In the case of antitrust concerns, it has at least one significant advantage over the current premerger antitrust policy regime. A tax can be applied after the merger is realized to be anti-competitive. The premerger policy requires the policymaker to predict if a potential merger will be noncompetitive. A prediction that is difficult to make.

Consider a tax $\bar{\tau}$ on gross profits $\pi_{f,i,t}$ higher than a threshold level $\bar{\pi}$. The gross-profit term in (**??**) becomes $\pi_{f,i,t} - \tau\left(\pi_{f,i,t}\right)$ where

$$\tau\left(\pi_{f,i,t}\right) = \bar{\tau}\max\left\{0, \pi_{f,i,t} - \bar{\pi}\right\}. \tag{61}$$

[25]The expectation operator is required since a merger can result in two possibilities: the acquiring combines the two firms' products or it kills the acquired companies products. Each case results in a different level of post-merger output.

This tax policy has two free parameters: $\bar{\tau}$ and $\bar{\pi}$. These are selected by solving that optimal policy problem of picking the pair $(\bar{\tau}, \bar{\pi})$ that maximizes household welfare. The tax policy $\bar{\tau}$ is restricted to $[0, 1]$ and $\bar{\pi}$ is restricted between the median and maximum level of gross profits in the baseline model. Under this tax policy, the antitrust policymaker does not block any transaction. The result is setting $\tau = 1$ and $\bar{\pi}$ to a level that is at the 98th percentile of gross profits in the baseline model. This policy results in household welfare **xxx**% greater than the baseline model. This tax has the effect of zeroing out the profits of every monopoly firm as well as the most concentrated duopoly markets.

Focusing on the case where $\bar{\tau} = 1$, Figure **??** displays key economic aggregates over different choices of the cutoff $\bar{\pi}$.

# 6 Conclusion

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**Table 1:** Outcome of Second Reviews by HHI and ΔHHI

**Panel A.** Number of Second Reviews

| | | Change in HHI (ΔHHI) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 0-99 | 100-199 | 200-299 | 300-499 | 500-799 | 800-1199 | 1200-2499 | 2500+ |
| Post-Merger HHI | 0-1799 | 14 | 48 | 39 | 28 | 10 | 1 | | |
| | 1800-1999 | 4 | 9 | 11 | 16 | 17 | | | |
| | 2000-2399 | 3 | 7 | 15 | 44 | 44 | 4 | | |
| | 2400-2999 | 3 | 6 | 11 | 24 | 58 | 36 | | |
| | 3000-3999 | 4 | 5 | 7 | 14 | 39 | 92 | | |
| | 4000-4999 | | 4 | 2 | 6 | 14 | 22 | 71 | |
| | 5000-6999 | 1 | 6 | 10 | 9 | 19 | 23 | 165 | 52 |
| | 7000+ | | | 1 | 1 | 3 | 9 | 27 | 248 |

**Panel B.** Percent of Second Reviews Resulting in a Challenge

| | | Change in HHI (ΔHHI) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 0-99 | 100-199 | 200-299 | 300-499 | 500-799 | 800-1199 | 1200-2499 | 2500+ |
| Post-Merger HHI | 0-1799 | | 35 | 49 | 61 | 30 | | | |
| | 1800-1999 | | 56 | 45 | 75 | 71 | | | |
| | 2000-2399 | 33 | 14 | 47 | 57 | 73 | 50 | | |
| | 2400-2999 | 33 | 67 | 55 | 75 | 76 | 72 | | |
| | 3000-3999 | 25 | 60 | 71 | 64 | 64 | 77 | 74 | |
| | 4000-4999 | | 50 | 50 | 83 | 71 | 82 | 96 | |
| | 5000-6999 | 100 | 100 | 80 | 89 | 100 | 91 | 88 | 90 |
| | 7000+ | | | 100 | 100 | 100 | 100 | 96 | 99 |

Note: The data is retrieved from *Horizontal Merger Investigation Data, Fiscal Years 1996-2011.* A blank value indicates a zero.

**Table 2:** Estimated Parameters and Target Moments

**Panel A.** Estimated Parameters

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| $\alpha$ | 0.65 | $\varphi_j$ | 0.22 |
| $\beta$ | 0.56 | $\varphi_x$ | 0.65 |
| $\eta$ | 0.52 | $\theta$ | 6.54 |
| $\lambda$ | 0.07 | $\zeta$ | 1.16 |
| $m$ | 2.12 | $z_1$ | 0.80 |
| $\omega$ | 26.58 | $z_3$ | 1.27 |
| $\varphi_a$ | 0.05 | $\bar{s}$ | 0.82 |

**Panel B.** Target Moments

| Description | Data | Model | Description | Data | Model |
|---|---|---|---|---|---|
| GDP Growth Rate | 2.41 | 2.39 | M&A-GDP Ratio | 0.05 | 0.05 |
| R&D-GDP Ratio | 0.04 | 0.04 | Avg. Merger Premium | 41.56 | 42.33 |
| Log Regression Sales | 1.39 | 1.05 | S.D. Merger Premium | 44.90 | 43.54 |
| Log Regression Sales$^2$ | $-1.11$ | $-1.12$ | Killer Acquisition Rate | 5.30 | 5.18 |
| Avg. Markup | 34.98 | 35.13 | Second Request Rate | 3.12 | 3.25 |
| S.D. Markup | 34.60 | 34.48 | Bottom 50% Sales Share | 1.40 | 1.23 |
| Avg. Firm Profitability | 0.15 | 0.15 | Top 10% Sales Share | 80.37 | 80.25 |

**Table 3:** Entry Effort By Firm Productivity

| Productivity | Baseline | No M&A | Difference |
|---|---|---|---|
| $z_1$ | 0.15 | 0.07 | -53% |
| $z_2$ | 0.24 | 0.21 | -13% |
| $z_3$ | 0.36 | 0.33 | -8% |

Note: The column "Baseline" is the entry rate in the baseline model. The column "No M&A" is the entry where M&A is shut off. The "Difference" column is the difference between the no M&A case and the baseline model.

**Table 4:** Sources of Aggregate Growth (Percent)

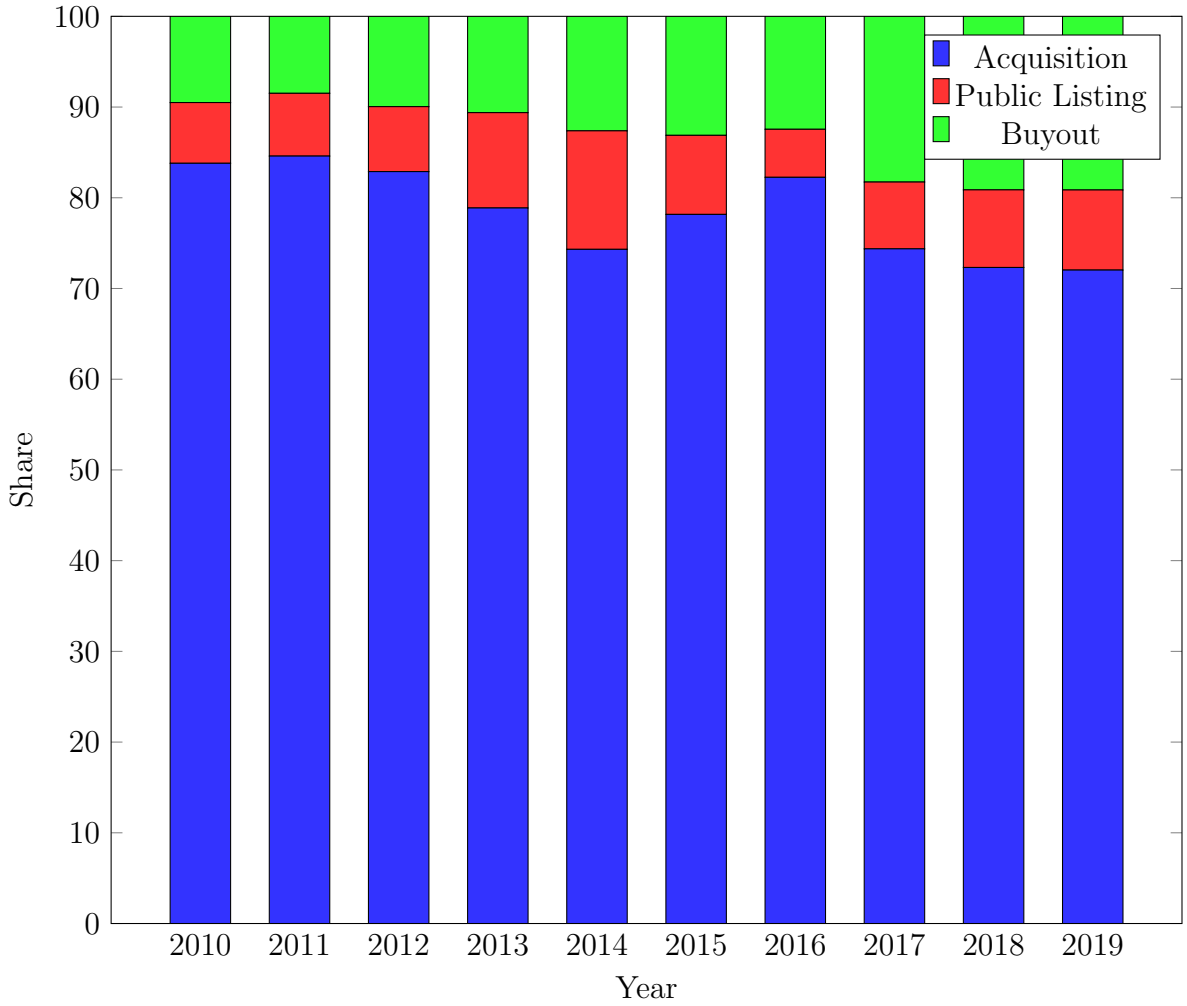
| Source | Baseline | No M&A | Difference |
|---|---|---|---|
| Total | 2.39 | 1.90 | -0.49 |
| Improvement | 0.69 | 0.95 | 0.26 |
| Entry | 1.29 | 0.94 | -0.35 |
| M&A Combine | 0.41 | 0.00 | -0.41 |
| M&A Kill | -0.01 | 0.00 | 0.01 |

Note: The column "Baseline" is the decomposition of the aggregate sources of growth in the baseline model. The column "No M&A" is the decomposition where M&A is shut off. The "Difference" column is the difference between the no M&A case and the baseline model.

**Table 5:** Alternative Antitrust Policies

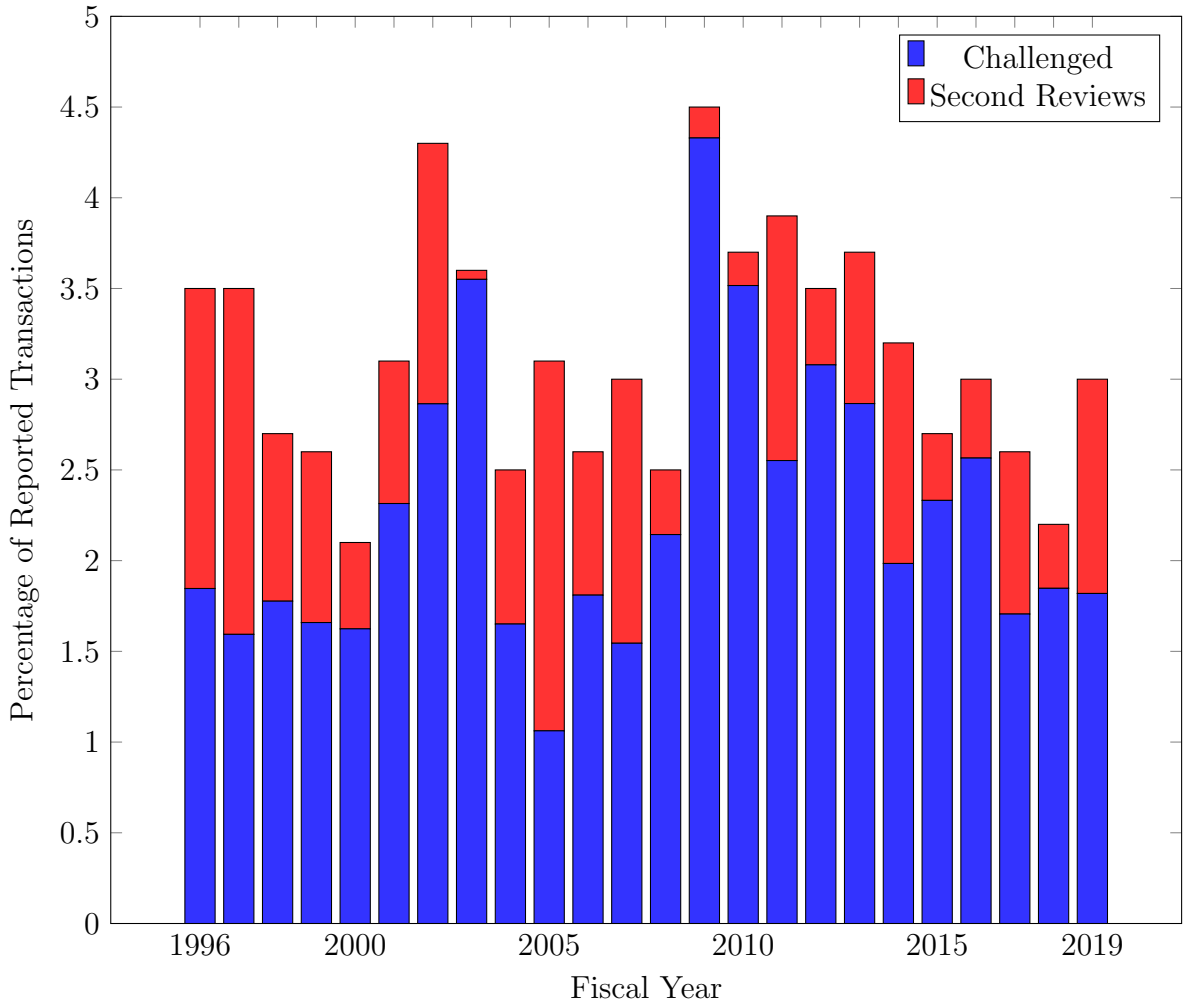
| | Baseline | No M&A | No Enforcement |
|---|---|---|---|
| **Welfare** | | | |
| Welfare Difference (%) | | -3.87 | -4.10 |
| $Y$ | 0.36 | 0.40 | 0.31 |
| $g$ | 2.39 | 1.90 | 2.96 |
| **Components of $Y$** | | | |
| $Z$ | 1.27 | 1.23 | 1.27 |
| $\mathcal{M}$ | 0.92 | 0.98 | 0.84 |
| $H$ | 0.31 | 0.33 | 0.29 |
| **Sources of Misallocation** | | | |
| Mean Markup(%) | 35.13 | 23.32 | 81.28 |
| S.D. Markup(%) | 34.48 | 4.75 | 72.56 |

**Figure 1:** Venture Capital Exits By Type of Exit




Note: This data was retrieved from the Q3 2022 Pitchbook-NVCA Venture Monitor.

**Figure 2:** Frequency of Second Antitrust Reviews and Challenges




Note: The percentage of transactions reported as part of the Premerger Notification Process which received second review and some form of challenge by the antitrust authority. The data is retrieved from the *Annual Reports to Congress Pursuant to the Hart-Scott-Rodino Antitrust Improvements Act of 1976* fiscal years 1996-2019. The report's time period is the government Oct. 1st to Sept. 31st fiscal year instead of the calendar year.

# A Derivations

## A.1 Aggregate Output

For firm $f$ in market $i$, their price is

$$p_{f,i,t} = m_{f,i,t} \frac{w_t}{z_f q_{f,i,t}}.$$

The price the fringe firms set

$$p_{c,i,t} = \frac{w_t}{\zeta z_{i,t} q_{i,t}}.$$

This means market $i$'s price index $p_{i,t}$ is given by

$$p_{i,t}^{1-\theta} = \frac{1}{|F_{i,t}|+1}\left(\sum_{f\in F_{i,t}} m_{f,i,t}^{1-\theta}(z_f q_{f,i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}\right) w_t^{1-\theta}.$$

Plugging this into the firm's and fringe's demand curve gives

$$\begin{aligned} y_{f,i,t} &= \frac{m_{f,i,t}^{-\theta}(z_f q_{f,i,t})^{\theta}}{\sum_{f'\in F_{i,t}} m_{f',i,t}^{1-\theta}(z_{f'} q_{f',i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}} \frac{P_t}{w_t} Y_t \\ y_{c,i,t} &= \frac{(\zeta z_{i,t} q_{i,t})^{\theta}}{\sum_{f\in F_{i,t}} m_{f,i,t}^{1-\theta}(z_f q_{f,i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}} \frac{P_t}{w_t} Y_t \end{aligned}$$

Firm $f$'s labor demand is

$$\begin{aligned} h_{f,i,t} &= \frac{y_{f,i,t}}{z_f q_{f,i,t}} \\ &= \frac{m_{f,i,t}^{-\theta}(z_f q_{f,i,t})^{\theta-1}}{\sum_{f'\in F_{i,t}} m_{f',i,t}^{1-\theta}(z_{f'} q_{f',i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}} \frac{P_t}{w_t} Y_t. \end{aligned}$$

The fringe's labor demand is

$$\begin{aligned} h_{c,i,t} &= \frac{y_{c,i,t}}{z_{i,t} q_{i,t}} \\ &= \frac{(\zeta z_{i,t} q_{i,t})^{\theta-1}}{\sum_{f\in F_{i,t}} m_{f,i,t}^{1-\theta}(z_f q_{f,i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}} \frac{P_t}{w_t} Y_t. \end{aligned}$$

Total labor demand in market $i$ is

$$\begin{aligned} h_{i,t} &= \sum_{f\in F_{i,t}} h_{f,i,t} + h_{c,i,t} \\ &= \frac{\sum_{f\in F_{i,t}} m_{f,i,t}^{-\theta}(z_f q_{f,i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}}{\sum_{f\in F_{i,t}} m_{f,i,t}^{1-\theta}(z_f q_{f,i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}} \frac{P_t}{w_t} Y_t. \end{aligned}$$

Converting the productivities into their relative form, we get

$$h_{i,t} = \frac{\sum_{f\in F_{i,t}} m_{f,i,t}^{-\theta}(\tilde{z}_{f,i,t}\tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}}{\sum_{f\in F_{i,t}} m_{f,i,t}^{1-\theta}(\tilde{z}_{f,i,t}\tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}} \frac{P_t}{w_t} Y_t.$$

Aggregating labor demand across markets, we see that

$$H_t = \int_0^1 \frac{\sum_{f\in F_{i,t}} m_{f,i,t}^{-\theta}(\tilde{z}_{f,i,t}\tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}}{\sum_{f\in F_{i,t}} m_{f,i,t}^{1-\theta}(\tilde{z}_{f,i,t}\tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}} \frac{P_t}{w_t} Y_t \,\mathrm{d}i.$$

Since aggregate labor supply is $H_t = 1$, we get that fraction of aggregate output compensating labor is

$$\frac{w_t}{P_t Y_t} = \int_0^1 \frac{\sum_{f \in F_{i,t}} m_{f,i,t}^{-\theta} (\tilde{z}_{f,i,t} \tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}}{\sum_{f \in F_{i,t}} m_{f,i,t}^{1-\theta} (\tilde{z}_{f,i,t} \tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}} \, \mathrm{d}i.$$

Returning to output, market $i$'s output is

$$\begin{aligned} y_{i,t} &= \left( \frac{1}{|F_{i,t}| + 1} \right)^{\frac{1}{\theta-1}} \left( \sum_{f \in F_{i,t}} y_{f,i,t}^{\frac{\theta-1}{\theta}} + y_{c,i,t}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}} \\ &= \left( \frac{\sum_{f \in F_{i,t}} m_{f,i,t}^{1-\theta} (z_f q_{f,i,t})^{\theta-1} + (\zeta z_{i,t} q_{i,t})^{\theta-1}}{|F_{i,t}| + 1} \right)^{\frac{1}{\theta-1}} \frac{P_t Y_t}{w_t} \\ &= z_{i,t} q_{i,t} \left( \frac{\sum_{f \in F_{i,t}} m_{f,i,t}^{1-\theta} (\tilde{z}_{f,i,t} \tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}}{|F_{i,t}| + 1} \right)^{\frac{1}{\theta-1}} \frac{P_t Y_t}{w_t}. \end{aligned}$$

It follows that aggregate output $Y_t$ is defined by

$$\log Y_t = \int_0^1 \log z_{i,t} + \log q_{i,t} + \frac{1}{\theta - 1} \log \left( \frac{\sum_{f \in F_{i,t}} m_{f,i,t}^{1-\theta} (\tilde{z}_{f,i,t} \tilde{q}_{f,i,t})^{\theta-1} + \zeta^{\theta-1}}{|F_{i,t}| + 1} \right) + \log \frac{P_t Y_t}{w_t} \, \mathrm{d}i.$$